

RAHUL KP

Software Engineer | AI/ML Engineer

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SUMMARY

Computer Science postgraduate and software engineer focused on building reliable, scalable, and production-oriented software and AI/ML systems. Strong foundation in **data structures and algorithms, object-oriented programming, software engineering, databases, distributed systems, machine learning, and backend development**. Experienced through research and personal projects in designing, implementing, testing, benchmarking, deploying, and documenting end-to-end systems using **Python, Java, C/C++, JavaScript, PyTorch, FastAPI, PostgreSQL, Redis, Docker, Kubernetes**, and cloud/infrastructure tooling.

Strong interest in solving engineering problems across **platform, backend, data, AI/ML, and product systems**, with an emphasis on correctness, maintainability, performance, testing, observability, and measurable outcomes.

EDUCATION

University of Calicut

Master of Science in Computer Science | 2024 – 2026 Kerala, India | ~77%

Relevant foundations: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Operating Systems, Computer Networks, Software Engineering, Artificial Intelligence, Machine Learning, Parallel Processing, Advanced Computing.

University of Calicut

Bachelor of Science in Computer Science | 2020 – 2023 Kerala, India

TECHNICAL SKILLS

Programming: Python, Java, C, C++, JavaScript, SQL

Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Algorithm Design, Complexity Analysis, Debugging, Software Design, Multithreading/Parallel Processing, Operating Systems, Computer Networks, Database Systems

Backend & APIs: FastAPI, Flask, Node.js, REST APIs, PostgreSQL, Redis

Frontend: React, Next.js, HTML, CSS, Tailwind CSS

AI / Machine Learning: PyTorch, NumPy, scikit-learn, XGBoost, LightGBM, Random Forest, Transformers, NLP, Computer Vision, Recommender Systems, Deep Learning

Generative AI: LLMs, RAG, LangChain, Hugging Face Transformers, LoRA, QLoRA, LLM Fine-tuning

Engineering & Infrastructure: Git, GitHub, GitHub Actions, Docker, Kubernetes, Nginx, Terraform, Prometheus, Grafana, Linux, Tailscale

Databases & Data: PostgreSQL, Redis, SQL, relational data modeling, query optimization

Engineering Practices: Unit Testing, Integration Testing, API Testing, CI/CD, Monitoring, Documentation, Benchmarking, Performance Analysis, Production Deployment

SELECTED PROJECTS

Neural Collaborative Filtering Recommendation System

Python · NumPy · PyTorch · FastAPI · PostgreSQL · Next.js · Docker · Kubernetes

- Designed and implemented a complete **Neural Collaborative Filtering (NCF)** recommendation system covering mathematical formulation, model implementation, evaluation, API serving, frontend integration, and production architecture.
- Implemented the core recommendation models **from scratch using NumPy**, including Generalized Matrix Factorization (GMF) and Multi-Layer Perceptron (MLP), before reproducing the models using PyTorch.
- Built a reproducible MovieLens 1M evaluation pipeline using **1:4 negative sampling** and leave-one-out evaluation with 99 negative candidates.
- Evaluated recommendation quality using **Hit@10 and NDCG@10**, achieving approximately **0.628 Hit@10** and demonstrating comparable recommendation quality between the scratch and PyTorch implementations.

- Benchmarked the implementations and measured approximately **12.8× epoch-level training speedup with PyTorch** over the NumPy implementation.
- Built a production-oriented architecture using **FastAPI model serving, PostgreSQL, Next.js, Docker, Nginx, Kubernetes, monitoring, and infrastructure-as-code concepts**.
- Implemented approximately **1.23M trainable parameters** and documented model architecture, training methodology, evaluation methodology, benchmarking, and deployment design.
- Authored a research preprint documenting the implementation, empirical evaluation, and production architecture.

Engineering focus: algorithms, numerical computing, benchmarking, APIs, database integration, deployment, reproducibility, performance.

PhishGuard — Machine Learning Phishing Detection System

Python · scikit-learn · Machine Learning · Testing · REST API

- Developed a machine-learning-based phishing detection system for classifying potentially malicious URLs.
- Built the complete pipeline covering **data preprocessing, feature engineering, model training, validation, evaluation, and inference**.
- Achieved approximately **95% classification accuracy** and **~0.98 ROC-AUC** on the evaluated dataset.
- Developed automated tests covering **222 test cases** with approximately **98.7% test coverage**, emphasizing regression resistance and maintainability.
- Exposed model functionality through an application/API layer for practical inference.
- Applied systematic evaluation and debugging to identify model and pipeline failures before deployment.

Engineering focus: ML engineering, testing, validation, reliability, API integration.

TaskFlow — Full-Stack Task Management Platform

JavaScript · React · Node.js · Express · MongoDB

- Developed a full-stack task management application using a modern JavaScript stack.

- Designed RESTful backend APIs and integrated frontend workflows for task creation, modification, tracking, and management.
- Implemented application-level validation, authentication-related workflows, database interaction, and modular frontend/backend components.
- Applied software engineering practices across development, debugging, integration, and deployment.

Engineering focus: full-stack development, REST APIs, databases, application architecture.

ANN From Scratch

Python · NumPy · Mathematics · Neural Networks

- Implemented an artificial neural network from first principles using NumPy without relying on high-level deep-learning training abstractions.
- Implemented forward propagation, activation functions, loss computation, backpropagation, gradient calculation, and parameter updates.
- Used mathematical derivations to verify gradient flow and understand optimization behavior.
- Benchmarked the implementation against framework-based approaches to understand the trade-offs between low-level control and optimized execution.

Engineering focus: algorithms, numerical computing, mathematical reasoning, debugging.

CNN Image Classification

Python · NumPy · PyTorch · Computer Vision

- Built a convolutional neural network pipeline for handwritten digit classification.
- Implemented and evaluated convolutional architectures and training workflows.
- Achieved approximately **98.6% classification accuracy** on the evaluated MNIST task.
- Investigated model behavior through validation metrics and iterative experimentation.

Engineering focus: computer vision, deep learning, experimentation, performance evaluation.

Enterprise RAG / LLM Application

Python · FastAPI · LangChain · Hugging Face · PostgreSQL · Redis

- Designed an end-to-end retrieval-augmented generation architecture for enterprise knowledge retrieval and question answering.
- Implemented document ingestion, chunking, embeddings/retrieval, context construction, LLM inference, and API serving.
- Used PostgreSQL/Redis-oriented architecture for persistent data and application-level caching/state management.
- Structured the system for production deployment with API separation, containerization, and observability considerations.

Engineering focus: backend architecture, information retrieval, LLM systems, APIs, data pipelines.

RESEARCH & PUBLICATIONS

Neural Collaborative Filtering Recommendation System

Researcher / Author

Design of a Neural Collaborative Filtering-Based Movie Recommendation System: From-Scratch Implementation, PyTorch Benchmarking, and Production Architecture

- Investigated neural collaborative filtering through mathematical formulation, from-scratch implementation, framework reproduction, empirical benchmarking, and production system design.
 - Compared NumPy and PyTorch implementations under consistent experimental conditions.
 - Evaluated ranking quality using **Hit@10** and **NDCG@10**.
 - Documented engineering trade-offs between educational from-scratch implementations and optimized deep-learning frameworks.
 - Preprint DOI: **10.20944/preprints202605.0449.v1**
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CERTIFICATIONS

IBM — Generative AI Engineering Professional Certificate

Oracle — Generative AI Professional

ISRO / IIRS — AI/ML for Geodata Analytics

40+ additional technical certifications and learning credentials across software engineering, AI/ML, cloud, data, and related technologies.

ENGINEERING HIGHLIGHTS

- **End-to-end ownership:** comfortable moving from problem definition and algorithm design through implementation, testing, API integration, deployment, monitoring, and documentation.
 - **Fundamentals-first engineering:** strong preference for understanding mathematical and algorithmic foundations before using high-level frameworks.
 - **Performance-oriented:** experienced in benchmarking implementations and quantifying framework/runtime trade-offs.
 - **Production mindset:** familiar with containerization, CI/CD, databases, caching, reverse proxies, orchestration, infrastructure-as-code, and observability.
 - **Research-oriented:** experienced in reproducible experimentation, technical writing, empirical evaluation, and communicating engineering decisions.
 - **AI-assisted engineering:** experienced with modern AI/LLM development while emphasizing validation, testing, correctness, and human review.
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ADDITIONAL TECHNICAL EXPERIENCE

Software Engineering: SDLC, requirements analysis, system design, implementation, testing, debugging, code review, documentation, deployment

Algorithms: Arrays, Strings, Linked Lists, Stacks, Queues, Trees, Graphs, Searching, Sorting, Dynamic Programming, Recursion, Complexity Analysis

Machine Learning: Supervised Learning, Neural Networks, CNNs, Recommender Systems, Classification, Ranking, Model Evaluation, Feature Engineering

Systems: REST architecture, relational databases, caching, containerization, service deployment, monitoring, CI/CD

LINKS

Portfolio: <https://rahulkp-ai.github.io> **GitHub:** <https://github.com/rahulkp-ai> **LinkedIn:** <https://linkedin.com/in/rahulkp-ai> **Hugging Face:** <https://huggingface.co/rahulkp-ai> **Google Scholar:** <https://scholar.google.com> **ORCID:** <https://orcid.org/0009-0009-3403-6670>